PIK Structural Credit Pricing

How Much Extra Spread Should Investors Demand for PIK Risk?

Structural Credit + Monte Carlo Analysis

Agenda

1. **What Are PIK Bonds?**
   How cash-pay, full PIK, and PIK toggle structures work
2. **Modelling Framework**
   Merton structural model [1,2] and reduced-form hazard rate model [3,4]
3. **The Credit Spectrum**
   BB+ to CCC- — how credit quality shapes PIK risk
4. **The PIK Feedback Loop**
   Why deferred coupons create a self-reinforcing leverage spiral [5,6]
5. **The PIK Premium**
   How much extra spread should investors demand? (+58bp to +628bp)
6. **What Drives the Premium?**
   Sensitivity to coverage, volatility, and model assumptions
7. **Borrower Optionality**
   Toggle exercise and its cost to lenders [8]
8. **Conclusions & Implications**
   Key findings for investors and risk managers

[1] Merton (1974) [2] Black & Cox (1976) [3] Duffie & Singleton (1999) [4] Jarrow & Turnbull (1995) [5] Leland (1994) [6] Collin-Dufresne & Goldstein (2001) [7] Glasserman (2003) [8] Longstaff & Schwartz (2001) — see References slide

What Are PIK Bonds?

PIK bonds defer cash coupons into notional growth. If the issuer defaults while notional has grown, investors lose more. The question: how much extra spread compensates for this risk?

Stylized 5-year 8.5% semi-annual fixed-coupon bond (4.5% risk-free + 400bp credit spread)

Structure	Coupon Mechanics	PIK Schedule
Cash-Pay	All coupons paid in cash at each date	None
Full PIK	All coupons accreted to notional	Full PIK
PIK Toggle	Borrower decides cash vs PIK	State-dependent

Key Parameters

- Risk-free rate: 4.5%
- Coupon spread: 400bp (total coupon = 8.5%)
- Maturity: 5 years
- Notional: 100
- Monte Carlo paths: 25,000
- Antithetic variates + Brownian bridge [7]
- Seed: 42 (reproducible)

PIK Mechanics in Detail

- Cash-Pay: investor receives coupon cash each period. Notional stays at par.
- Full PIK: no cash coupons — each coupon accretes to notional. At maturity, $N(T) = 100 \times (1 + 4.25\%)^{10} = 151.6$
- PIK Toggle: borrower chooses each period whether to pay cash or PIK. Creates adverse selection risk [8].

[7] Glasserman (2003), Monte Carlo Methods in Financial Engineering [8] Longstaff & Schwartz (2001), Valuing American Options by Simulation

Merton (1974) Structural Credit Model

The Merton model treats the company's equity as a call option on its assets. Default occurs when assets fall below the debt barrier. This structural approach naturally captures how PIK-inflated debt changes default dynamics — unlike hazard-rate models.

Asset Dynamics (GBM under risk-neutral measure) [1]:

$dV = (r - q) \cdot V \cdot dt + \sigma v \cdot V \cdot dW$

Distance-to-Default:

$DD = [\ln(V/B) + (r - q - \frac{1}{2}\sigma^2) \cdot T] / (\sigma \cdot \sqrt{T})$

Default Probability:

$PD(T) = N(-DD)$ where $N(\cdot)$ is the standard normal CDF

Implied Credit Spread:

$s = -\ln(1 - PD \cdot LGD) / T$, $LGD = 1 - R$

Terminal vs First-Passage [2]:

- Terminal: default only at maturity (Merton 1974)
- First-passage: default at any time $V < B$ (Black & Cox 1976)
- First-passage defaults are 1.5-2x higher than terminal rates

Analytical Credit Metrics

Issuer	Barrier	DD	PD(5Y)	Impl Spread
BB+	136.8	1.13	12.94%	134 bp
BB	132.5	1.08	13.90%	159 bp
BB-	107.3	1.05	14.70%	185 bp
B+	106.8	0.95	16.99%	234 bp
B	84.3	0.86	19.51%	282 bp
B-	70.0	0.69	24.60%	378 bp
CCC+	74.7	0.50	30.76%	525 bp
CCC	69.6	0.37	35.75%	674 bp
CCC-	81.3	0.14	44.32%	946 bp

Barriers calibrated to match market-implied PD via from_target_pd (terminal basis)

[1] Merton, R.C. (1974), On the Pricing of Corporate Debt, Journal of Finance 29(2) [2] Black, F. & Cox, J.C. (1976), Valuing Corporate Securities, Journal of Finance 31(2)

Reduced-Form Hazard Rate Model

The hazard rate model prices bonds using a constant default intensity λ calibrated to market spreads. It captures the timing and notional effects of PIK but cannot capture the leverage feedback loop — it serves as our baseline for measuring what the structural model adds.

Model Formulation

Survival function:

$S(t) = \exp(-\lambda \cdot t)$

Bond pricing with default [3,4]:

$$PV = \Sigma\, c_i \cdot D(t_i) \cdot S(t_i) + N \cdot D(T) \cdot S(T) + R \cdot N \cdot \Sigma\, D(t_i) \cdot [S(t_{n-1}) - S(t_i)]$$

Credit triangle approximation:

$\lambda_0 \approx s / (1 - R)$

- λ is FLAT — does not respond to leverage changes
- Captures PIK timing: PIK shifts notional to maturity
- Misses feedback: PIK → higher leverage → higher λ

Two Models, Two Purposes

Hazard Rate (Reduced-Form):

- Constant λ calibrated from market
- Tractable, closed-form pricing
- PIK premium from timing effect only

Merton MC (Structural + Feedback):

- Asset dynamics drive default
- Endogenous λ(L) rises with leverage [5]
- Dynamic recovery R(N) falls with PIK accrual
- Captures the feedback loop

Gap between models = value of the feedback loop

The difference between the two models isolates the structural feedback premium:

At CCC-: HR premium = +220bp, MC premium = +628bp → feedback adds +408bp (≈ 2/3 of total PIK premium)

[3] Duffie, D. & Singleton, K.J. (1999), Modeling Term Structures of Defaultable Bonds, RFS 12(4) [4] Jarrow, R.A. & Turnbull, S.M. (1995), Pricing Derivatives on Financial Securities Subject to Credit Risk, JoF 50(1)

The PIK Feedback Loop

PIK creates a vicious cycle: unpaid coupons inflate the debt, pushing leverage higher, which raises default probability AND lowers recovery. This self-reinforcing spiral is the core mechanism driving the PIK premium.

Endogenous Hazard (Power-Law) [5,6]

$\lambda(L) = \lambda_0 \cdot (L / L_0)^{\beta}$

- λ_0 = base hazard, calibrated from market spreads
- L_0 = base leverage at origination (B/V)
- β = exponent controlling feedback speed

As PIK inflates notional and leverage rises, hazard accelerates non-linearly.

Dynamic Recovery (Floored Inverse)

$R(N) = \max(\text{floor}, R_0 \cdot N_0 / N)$

- R_0 = base recovery at par notional N_0
- floor = minimum recovery rate (10%)
- α variant: $R(N) = R_0 \cdot (N_0/N)^{\alpha}$

PIK-inflated notional means recovery < base rate.

Primary driver of full-PIK premium.

Feedback Demonstration: B+ Issuer (barrier = 106.8, base leverage = 0.63)

LTV	Hazard	Notional	Recovery
0.50	2.36%	79.6	35.00%
0.70	4.62%	111.4	31.41%
0.90	7.64%	143.3	24.43%
1.00	9.44%	159.2	21.99%
1.10	11.42%	175.1	19.99%

[5] Leland, H.E. (1994), Corporate Debt Value, Bond Covenants, and Optimal Capital Structure, JoF 49(4) [6] Collin-Dufresne, P. & Goldstein, R.S. (2001), Do Credit Spreads Reflect Stationary Leverage Ratios?, JoF 56(5)

The Credit Spectrum

We test PIK economics across 9 issuers from BB+ to CCC-. The same PIK structure behaves very differently across credit quality — a BB+ PIK barely matters, but a CCC- PIK premium can exceed 600bp.

Issuer	Assets (V)	Asset Vol (σ)	Mkt Spread	h_0 (cal)	h≈s/(1-R)	R_0
BB+	200	20%	150bp	2.77%	3.00%	50%
BB	190	20%	175bp	2.99%	3.18%	45%
BB-	180	25%	200bp	3.18%	3.33%	40%
B+	170	25%	250bp	3.72%	3.85%	35%
B	150	30%	300bp	4.34%	4.44%	32%
B-	130	35%	400bp	5.65%	5.71%	30%
CCC+	120	35%	550bp	7.35%	7.33%	25%
CCC	115	40%	700bp	8.85%	8.75%	20%
CCC-	110	40%	975bp	11.71%	11.47%	15%

Market-Implied Default Probabilities: PD(t) = 1 - exp(-λt)

Issuer	λ (bp)	PD(1Y)	PD(3Y)	PD(5Y)	PD(7Y)	PD(10Y)
BB+	277	2.73%	7.98%	12.94%	17.63%	24.20%
B	434	4.25%	12.21%	19.51%	26.20%	35.21%
CCC-	1171	11.05%	29.63%	44.32%	55.95%	69.00%

Hazard rates calibrated via reduced-form model [3,4]. Credit triangle approximation: λ ≈ s / (1 - R), see Duffie & Singleton (1999).

The PIK Premium: How Much Extra Spread?

After calibrating to market, investors should demand +58bp (BB+) to +628bp (CCC-) additional spread for full PIK structures. The premium scales exponentially with credit risk.

Issuer	Mkt Spread	Cash* Z	PIK* Z	Toggle* Z	PIK-Cash*	Toggle-Cash*
BB+	150bp	149bp	206bp	150bp	+58bp	+1bp
BB	175bp	174bp	241bp	176bp	+68bp	+2bp
BB-	200bp	199bp	275bp	228bp	+76bp	+29bp
B+	250bp	253bp	358bp	295bp	+105bp	+43bp
B	300bp	302bp	438bp	424bp	+136bp	+122bp
B-	400bp	407bp	616bp	621bp	+209bp	+214bp
CCC+	550bp	548bp	848bp	845bp	+300bp	+297bp
CCC	700bp	698bp	1096bp	1095bp	+398bp	+397bp
CCC-	975bp	981bp	1609bp	1586bp	+628bp	+606bp

* = MC-calibrated: cash Z-spread matches market by construction

Calibrated PIK premiums are wider: +628bp at CCC- (vs +409bp uncalibrated)

Uncalibrated barriers inflate cash-pay spreads disproportionately — see Appendix H

Z-spreads computed on cash-equivalent basis for cross-structure comparability. MC calibration: bisect on barrier to match cash Z-spread to market. See [7] for MC methodology.

Why Simple Models Underestimate PIK Risk

Traditional hazard-rate models [3,4] miss 2/3 of the PIK premium for stressed issuers. A CCC- PIK bond is mispriced by ~400bp if you ignore the leverage spiral [5,6]. Structural models with feedback are essential for accurate pricing.

Issuer	λ (bp)	HR Cash Z	HR PIK Z	HR PIK-Cash	MC Cash* Z	MC PIK* Z	MC PIK-Cash*
BB+	277	149bp	197bp	+48bp	149bp	206bp	+58bp
BB	299	174bp	227bp	+53bp	174bp	241bp	+68bp
BB-	318	199bp	257bp	+58bp	199bp	275bp	+76bp
B+	372	249bp	316bp	+67bp	253bp	358bp	+105bp
B	434	299bp	375bp	+76bp	302bp	438bp	+136bp
B-	565	398bp	493bp	+95bp	407bp	616bp	+209bp
CCC+	735	548bp	673bp	+125bp	548bp	848bp	+300bp
CCC	885	698bp	855bp	+157bp	698bp	1096bp	+398bp
CCC-	1171	973bp	1193bp	+220bp	981bp	1609bp	+628bp

At CCC-: Hazard-rate PIK premium = +220bp vs Structural MC = +628bp

The structural feedback (endogenous hazard + dynamic recovery) accounts for nearly 2/3 of the PIK premium

HR = reduced-form hazard rate model [3,4] with flat λ. MC = Merton structural model [1,2] with endogenous hazard [5] and dynamic recovery, calibrated to market.

What Drives the Premium: Asset Coverage

Asset coverage is the single most important driver. The premium drops from +372bp at 1.1x coverage to just +26bp at 2.15x. Better collateral dramatically reduces PIK risk.

Sweep V from 110 to 220 (coverage 1.10x to 2.15x), fixed barrier at par, σ=30%, h_0=6%

Coverage	Cash Price	PIK Price	Discount	Cash Z	PIK Z	PIK Premium
1.10x	92.98	79.81	+13.17	557bp	929bp	+372bp
1.25x	97.05	86.67	+10.38	454bp	728bp	+273bp
1.40x	100.42	92.34	+8.08	372bp	574bp	+202bp
1.55x	103.33	97.23	+6.10	304bp	450bp	+146bp
1.70x	105.66	101.16	+4.50	251bp	355bp	+104bp
1.85x	107.64	104.49	+3.15	207bp	277bp	+71bp
2.00x	109.19	107.10	+2.09	173bp	219bp	+46bp
2.15x	110.48	109.28	+1.20	145bp	171bp	+26bp

Key Insight: PIK premium is strongly non-linear in coverage

- At 1.10x coverage: +372bp PIK premium (asset barely covers debt)
- At 2.15x coverage: +26bp PIK premium (defaults are rare, PIK accrual harmless)
- The relationship is convex — premiums accelerate as coverage drops below 1.5x

Structural framework per Merton (1974) [1] with first-passage default [2] and endogenous hazard [5]. Coverage = V/B at origination.

What Drives the Premium: Asset Volatility

Higher volatility amplifies PIK risk exponentially — at 45% vol the PIK premium is 9x larger than at 20% vol. Volatile sectors (e.g. tech, energy) face outsized PIK costs.

Fixed 1.40x coverage (V=140, barrier=par), sweep σ from 20% to 45%

Vol (σ)	Cash Price	PIK Price	Discount	Cash Z	PIK Z	PIK Premium
20%	109.17	107.07	+2.10	173bp	219bp	+46bp
25%	104.58	99.34	+5.24	275bp	398bp	+123bp
30%	100.42	92.34	+8.08	372bp	574bp	+202bp
35%	96.90	86.42	+10.48	458bp	735bp	+277bp
40%	93.81	81.20	+12.61	536bp	887bp	+351bp
45%	91.06	76.58	+14.48	608bp	1031bp	+424bp

Volatility amplifies PIK risk through three channels:

1. More frequent barrier crossings → higher default rate [2]
2. More extreme PIK paths → wider leverage swings → deeper recovery erosion
3. Non-linear amplification → both default frequency AND loss severity worsen simultaneously

PIK premium at 20% vol: +46bp. At 45% vol: +424bp (×9.2x increase)

GBM asset dynamics [1] with first-passage barrier crossing [2] and Brownian bridge correction [7].

What Drives the Premium: Feedback Speed (β and α)

The speed of the feedback loop is a key modelling assumption. How fast hazard rises with leverage (β) and how fast recovery erodes with notional growth (α) can swing the premium by 100bp+. Getting these parameters right is critical.

Endogenous Hazard Speed (β) [5,6] → Drives Toggle Premium

Issuer	Metric	β=0	β=1	β=2	β=4
BB-	Toggle-Cash	+0bp	+2bp	+29bp	+66bp
B	Toggle-Cash	+0bp	+53bp	+122bp	+145bp
CCC	Toggle-Cash	+398bp	+393bp	+397bp	+399bp

Recovery Decay Speed (α) → Drives PIK Premium

Issuer	Metric	α=0	α=0.5	α=1.0	α=2.0
BB-	PIK-Cash	+16bp	+48bp	+76bp	+118bp
B	PIK-Cash	+66bp	+104bp	+136bp	+186bp
CCC	PIK-Cash	+294bp	+351bp	+398bp	+473bp

Two distinct feedback channels:

β (Hazard Speed) [5,6]:

- Controls how quickly hazard ramps as leverage rises
- Drives the toggle exercise decision (PIK when hazard is high)
- At β=0, toggle sees flat hazard and rarely PIKs; at β=4, toggle aggressively PIKs

α (Recovery Decay):

- Controls how fast recovery erodes as PIK inflates notional
- Directly affects loss severity — primary driver of full-PIK premium

[5] Leland (1994): endogenous default barrier and leverage-dependent hazard. [6] Collin-Dufresne & Goldstein (2001): stationary leverage ratios and credit spread dynamics.

- At α=0, recovery is constant; at α=2.0, recovery collapses quickly

Borrower Optionality: Toggle Strategies

Borrowers exercise the toggle at the worst time for lenders — they PIK when the company is most stressed, seeding the leverage spiral on already-distressed paths. Optimal exercise [8] produces spreads nearly as wide as full PIK (adverse selection).

B+ Issuer Results

Strategy	Price	Z-Spread	E[Loss]	Def Rate	PIK%	Terminal Notional
Cash-Pay	108.21	194bp	7.89%	16.77%	0.0%	100.0
Full PIK	105.46	255bp	10.24%	16.77%	100.0%	151.6
Threshold (h > 10%)	106.51	232bp	9.34%	16.77%	6.6%	100.9
Stochastic (sigmoid)	106.59	230bp	9.27%	16.77%	15.5%	105.2
Optimal (nested MC) [8]	105.99	244bp	9.78%	16.77%	14.6%	104.1

Toggle Exercise Strategies:

Threshold: Deterministic rule — PIK when hazard rate > 10%

Stochastic: Smooth sigmoid — $P(PIK) = \sigma(a + b \cdot x)$, avoids hard threshold discontinuity

Optimal: Forward-looking nested MC [8] — maximizes equity value (most expensive to compute)

Insight: Optimal exercise produces spreads close to full PIK (244bp vs 255bp)

The borrower rationally chooses PIK when it's worst for the lender (adverse selection)

[8] Longstaff, F.A. & Schwartz, E.S. (2001), Valuing American Options by Simulation: A Simple Least-Squares Approach, RFS 14(1). Adapted for toggle exercise as nested MC.

Conclusions (1/2)

Five key takeaways: PIK risk is non-linear, feedback-driven, and underestimated by standard models.

1 PIK premium is non-linear in LTV

BB+ (~50% LTV): PIK barely matters. CCC (~87% LTV): PIK premium can exceed 500bp.

2 The feedback loop drives PIK risk [5,6]

Endogenous hazard + dynamic recovery create a self-reinforcing spiral that dramatically worsens PIK economics.

3 Simple hazard rates miss the story [3,4]

Flat λ captures timing/notional effects but underestimates premium by 2-3x for stressed issuers. CCC-: +220bp (HR) vs +628bp (MC).

4 Toggle is not free [8]

Borrower PIKs precisely when credit deteriorates — negative convexity for the investor. Toggle premium sits between cash and full PIK.

5 Volatility amplifies PIK risk

Higher σ increases both default frequency AND loss severity [1,2]. At 40% vol, PIK premium >350bp; at 20% vol, only ~46bp.

Reference numbers: see References slide. [1] Merton (1974) [2] Black & Cox (1976) [3] Duffie & Singleton (1999) [5] Leland (1994)

Conclusions (2/2)

Model robustness, duration implications, and measurement considerations for practitioners.

6 MC-to-market calibration eliminates fudge factors

Calibrating the barrier directly to market Z-spread ensures cash base case matches by construction. PIK differentials are computed consistently.

7 Toggle can be worse than full PIK (adverse selection)

Toggle concentrates PIK on the worst paths, seeding the leverage spiral on already-stressed paths. Can produce larger investor losses than mandatory full PIK.

8 PIK bonds carry ~0.75yr longer modified duration

Full-PIK ModDur ~4.8yr vs ~4.1yr for cash-pay (~18% extension). Greater interest rate sensitivity and higher DV01 per dollar of notional.

9 Metrics use the right cashflow basis

Spread/yield metrics: cash-equivalent basis for cross-structure comparability. Risk metrics (duration, DV01): actual PIK cashflows for true rate sensitivity.

10 Calibration parameter choice does not affect PIK premiums

Full-PIK premium is identical under barrier or vol calibration. Toggle shows small differences. Model is robust to calibration method.

Key Takeaway

PIK risk is fundamentally a leverage spiral risk.
Structural models with endogenous hazard and dynamic recovery reveal premiums 2-3x larger than simple hazard-rate approaches, especially for stressed issuers where it matters most.

+58bp

BB+ PIK Premium

+628bp

CCC- PIK Premium

+0.75yr

Duration Extension

References

[1] **Merton, R.C. (1974)** "On the Pricing of Corporate Debt: The Risk Structure of Interest Rates." Journal of Finance, 29(2), 449-470.

[2] **Black, F. & Cox, J.C. (1976)** "Valuing Corporate Securities: Some Effects of Bond Indenture Provisions." Journal of Finance, 31(2), 351-367.

[3] **Duffie, D. & Singleton, K.J. (1999)** "Modeling Term Structures of Defaultable Bonds." Review of Financial Studies, 12(4), 687-720.

[4] **Jarrow, R.A. & Turnbull, S.M. (1995)** "Pricing Derivatives on Financial Securities Subject to Credit Risk." Journal of Finance, 50(1), 53-85.

[5] **Leland, H.E. (1994)** "Corporate Debt Value, Bond Covenants, and Optimal Capital Structure." Journal of Finance, 49(4), 1213-1252.

[6] **Collin-Dufresne, P. & Goldstein, R.S. (2001)** "Do Credit Spreads Reflect Stationary Leverage Ratios?" Journal of Finance, 56(5), 1929-1957.

[7] **Glasserman, P. (2003)** Monte Carlo Methods in Financial Engineering. Springer. (Brownian bridge barrier correction, antithetic variates, variance reduction.)

[8] **Longstaff, F.A. & Schwartz, E.S. (2001)** "Valuing American Options by Simulation: A Simple Least-Squares Approach." Review of Financial Studies, 14(1), 113-147.

[9] **Leland, H.E. & Toft, K.B. (1996)** "Optimal Capital Structure, Endogenous Bankruptcy, and the Term Structure of Credit Spreads." Journal of Finance, 51(3), 987-1019.

Appendix A: Monte Carlo Pricing Engine

The MC engine prices each bond by simulating 25,000 asset paths, checking for barrier crossings via Brownian bridge [7], processing coupons (cash or PIK), and computing Z-spreads on a cash-equivalent basis.

1. Asset Evolution

GBM [1]: $V(t+\Delta t) = V(t) \cdot \exp[(\mu - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t} \cdot Z]$

2. First-Passage Default

Brownian bridge barrier-crossing correction [2,7]. Crossing prob: $p = \exp(-2x_0x_1 / \sigma^2\Delta t)$

3. Coupon Processing

Cash → pay coupon; PIK → accrete to notional; Toggle → evaluate exercise model [8]

4. Default Recovery

R(N_current) via DynamicRecoverySpec, discounted to valuation date

5. Maturity Payoff

Surviving paths: discount terminal notional N(T). For PIK: $N(T) = N_0 \cdot (1 + c/f)^n$

Variance Reduction & Z-Spread Computation

- Antithetic variates [7]: negate normal draws for half the paths (halves variance)
- Common random numbers (CRN): per-path deterministic RNG for calibration consistency
- Z-spread solved as: $\Sigma CF_i \cdot D(t_i) \cdot \exp(-z \cdot t_i) = MC\ PV$ (cash-equivalent basis)

[1] Black & Scholes (1973) [2] Black & Cox (1976) [7] Glasserman (2003) [8] Longstaff-Schwartz (2001)

Appendix B: Detailed Results — BB+ to BB-

BB+

Structure	Price	Z-Spread	YTM	ModDur	E[Loss]	Def Rate	PIK%
Cash-Pay	111.91	115bp	5.73%	4.11	4.75%	12.88%	0.0%
Full PIK	110.62	142bp	6.01%	4.86	5.84%	12.88%	100.0%
PIK Toggle	111.85	116bp	5.74%	4.11	4.80%	12.88%	0.2%

BB

Structure	Price	Z-Spread	YTM	ModDur	E[Loss]	Def Rate	PIK%
Cash-Pay	110.92	136bp	5.95%	4.10	5.58%	13.86%	0.0%
Full PIK	109.27	171bp	6.31%	4.86	6.99%	13.86%	100.0%
PIK Toggle	110.82	138bp	5.97%	4.10	5.67%	13.86%	0.3%

BB-

Structure	Price	Z-Spread	YTM	ModDur	E[Loss]	Def Rate	PIK%
Cash-Pay	109.97	156bp	6.16%	4.10	6.39%	14.64%	0.0%
Full PIK	108.02	198bp	6.59%	4.85	8.05%	14.64%	100.0%
PIK Toggle	108.74	183bp	6.43%	4.08	7.45%	14.64%	4.5%

Appendix C: Detailed Results — B+ to B-

B+

Structure	Price	Z-Spread	YTM	ModDur	E[Loss]	Def Rate	PIK%
Cash-Pay	108.21	194bp	6.55%	4.08	7.89%	16.77%	0.0%
Full PIK	105.46	255bp	7.18%	4.84	10.24%	16.77%	100.0%
PIK Toggle	106.51	232bp	6.94%	4.07	9.34%	16.77%	6.6%

B

Structure	Price	Z-Spread	YTM	ModDur	E[Loss]	Def Rate	PIK%
Cash-Pay	106.39	235bp	6.97%	4.06	9.44%	19.36%	0.0%
Full PIK	102.66	320bp	7.85%	4.83	12.62%	19.36%	100.0%
PIK Toggle	102.32	327bp	7.93%	4.03	12.91%	19.36%	30.7%

B-

Structure	Price	Z-Spread	YTM	ModDur	E[Loss]	Def Rate	PIK%
Cash-Pay	102.97	312bp	7.78%	4.03	12.35%	24.48%	0.0%
Full PIK	97.30	448bp	9.19%	4.80	17.18%	24.48%	100.0%
PIK Toggle	96.73	462bp	9.34%	3.98	17.67%	24.48%	63.7%

Appendix D: Detailed Results — CCC+ to CCC-

CCC+

Structure	Price	Z-Spread	YTM	ModDur	E[Loss]	Def Rate	PIK%
Cash-Pay	97.98	431bp	9.01%	3.99	16.60%	30.82%	0.0%
Full PIK	89.91	638bp	11.19%	4.77	23.47%	30.82%	100.0%
PIK Toggle	89.59	647bp	11.28%	3.90	23.74%	30.82%	69.2%

CCC

Structure	Price	Z-Spread	YTM	ModDur	E[Loss]	Def Rate	PIK%
Cash-Pay	93.44	545bp	10.21%	3.94	20.47%	35.74%	0.0%
Full PIK	83.51	818bp	13.10%	4.73	28.92%	35.74%	100.0%
PIK Toggle	83.27	825bp	13.18%	3.83	29.12%	35.74%	79.3%

CCC-

Structure	Price	Z-Spread	YTM	ModDur	E[Loss]	Def Rate	PIK%
Cash-Pay	85.93	748bp	12.36%	3.86	26.85%	44.27%	0.0%
Full PIK	72.78	1158bp	16.75%	4.67	38.05%	44.27%	100.0%
PIK Toggle	72.83	1156bp	16.73%	3.70	38.01%	44.27%	78.3%

Appendix E: Modified Duration Across Structures

PIK bonds have ~0.75yr longer duration than cash-pay (~18% more rate sensitivity). This is because PIK concentrates all value at maturity, extending the weighted-average cashflow timing.

Issuer	Cash ModDur	PIK ModDur	Toggle ModDur	PIK - Cash	Toggle - Cash
BB+	4.11	4.86	4.11	+0.75	0.00
BB	4.10	4.86	4.10	+0.76	0.00
BB-	4.10	4.85	4.08	+0.75	-0.02
B+	4.08	4.84	4.07	+0.76	-0.01
B	4.06	4.83	4.03	+0.77	-0.03
B-	4.03	4.80	3.98	+0.77	-0.05
CCC+	3.99	4.77	3.90	+0.78	-0.09
CCC	3.94	4.73	3.83	+0.79	-0.11
CCC-	3.86	4.67	3.70	+0.81	-0.16

Key patterns:

1. PIK extends duration by ~0.75yr

All coupons accreted to maturity as terminal notional, concentrating cashflow at the longest point.

2. Duration gap is stable across ratings

BB+ to CCC-: PIK - Cash ranges from +0.75 to +0.81yr. Cashflow structure — not credit — drives the gap.

3. Toggle converges to cash at high quality

BB+: toggle barely fires (0.2%), so duration matches cash. CCC-: toggle fires 78%, but duration stays near cash since non-PIK periods revert to cash timing.

Hedging implication: PIK bonds have ~18% greater rate sensitivity per dollar of notional.

Appendix F: Calibration — Barrier vs Asset Vol

The PIK premium is robust to calibration method — whether we calibrate via barrier or volatility, the full-PIK premium is identical. This validates that our results are not an artifact of the calibration approach.

Issuer	Mkt	Param	B (cal)	σ (cal)	Cash Z	PIK Z	PIK-Cash	Toggle-Cash
BB+	150bp	barrier	147.2	20.0%	149bp	206bp	+58bp	+1bp
		vol	136.8	22.3%	149bp	206bp	+58bp	+3bp
B+	250bp	barrier	117.5	25.0%	253bp	358bp	+105bp	+43bp
		vol	106.8	28.1%	253bp	358bp	+105bp	+56bp
B-	400bp	barrier	81.6	35.0%	407bp	616bp	+209bp	+214bp
		vol	70.0	40.2%	407bp	616bp	+209bp	+214bp
CCC	700bp	barrier	84.9	40.0%	698bp	1096bp	+398bp	+397bp
		vol	69.6	47.8%	698bp	1096bp	+398bp	+397bp

Result: PIK premiums are identical regardless of calibration parameter

- Full-PIK spread premium is the same under barrier or vol calibration
- Toggle premiums show small differences (toggle exercise responds differently to barrier vs vol)
- This validates model robustness — PIK pricing is not an artifact of the calibration method

Calibration via bisection on either barrier B or asset volatility σ to match cash Z-spread to observed market spread. See [7] for MC methodology.